Competitiveness in American Seafood Requires Promoting Fisheries to be Resilient

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Abstract

Recent federal executive actions prioritize restoring American seafood competitiveness through deregulation and production targets, yet these policies overlook the fundamental threat posed by climate-driven natural hazards. We argue that true seafood competitiveness cannot be achieved through production gains alone; it requires treating disaster resilience as a form of industrial policy. We use the Gulf of Mexico fisheries as an example of a system increasingly exposed to natural hazards, the main of which are hurricanes. The Gulf generates approximately $910 million annually in commercial dockside revenue, supports over 250,000 jobs, and accounts for one-third of the nation's recreational fishing expenditures, all concentrated in states with the highest hurricane exposure in the nation. As such, hurricanes impose a double jeopardy on these fisheries by simultaneously degrading ecosystems and destroying physical infrastructure. And while federal institutions such as NOAA and FEMA coordinate post-disaster relief, these mechanisms remain reactive and administratively burdensome, leaving the sector vulnerable to compounding losses. This climate vulnerability is not unique to the Gulf nor to hurricanes. Other climate-driven impacts such as marine heatwaves and harmful algal blooms are becoming increasingly prevalent and impose analogous shocks in the Gulf, as well as on Atlantic and Pacific fisheries. We propose a three-pillar framework that encompasses adaptive governance, a politically insulated safety net, and quantified climate risk assessments as the foundation for a durable, competitive, and resilient American seafood industry.

Keywords: climate adaptation, disaster resilience, fisheries management, Gulf of Mexico, hurricane impacts, seafood policy

Background

Recent executive actions and federal directives in the U.S. have prioritized restoring American seafood competitiveness to ensure food security and economic growth [Executive Office of the President of the United States, 2025]. However, policies do not exist in a vacuum, and concurrent climate challenges, such as hurricanes and associated storms, may inadvertently undermine these goals. As climate change drives ocean and atmospheric temperatures, the frequency and intensity of North Atlantic hurricanes and named storms are increasing [Vecchi et al., 2021], placing the very regions essential to domestic seafood production in the crosshairs (Fig 1). Here, we analyze the intersection of seafood policy and hurricane disaster management, arguing that true competitiveness cannot be achieved without a comprehensive resilience strategy for the Gulf of Mexico (currently referred to as the Gulf of America by the U.S. government).

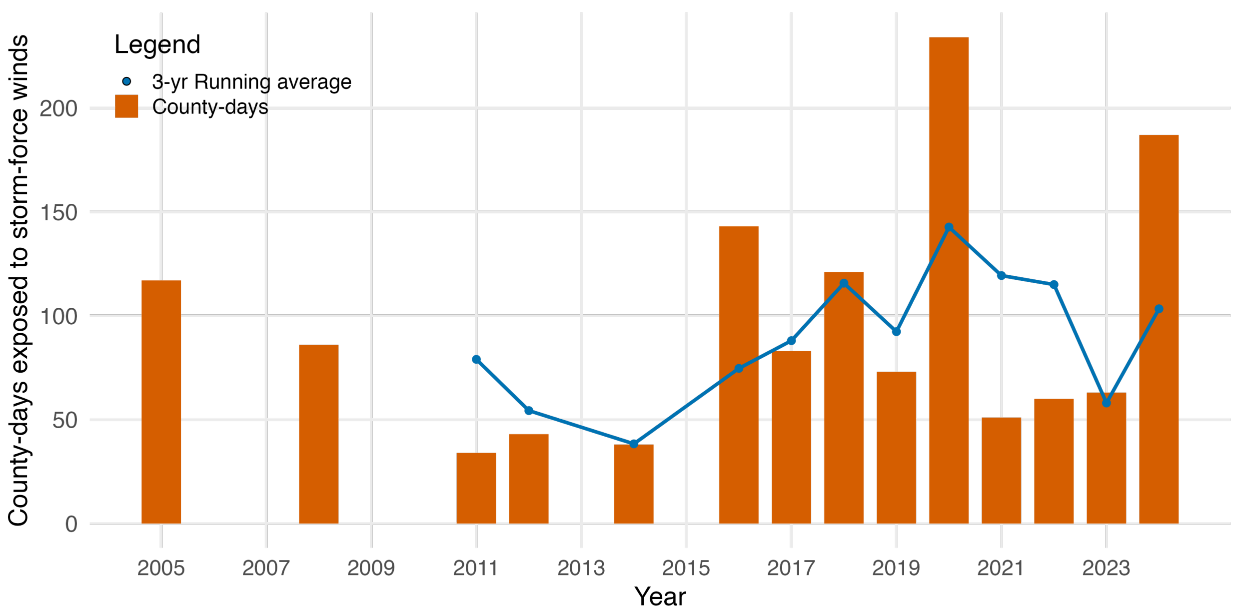

Figure 1: Fishing counties are increasingly exposed to storm-force winds. The horizontal axis shows time in years. The vertical axis shows exposure to storm-force winds (maximum sustained winds > 18 m/s), measured as the sum of days and counties exposed in a given year. Bars show annual exposure and lines show the 3-yr running average of exposure. County-level exposure data come from Molina and Rudik [forthcoming].

The Gulf of Mexico is the backbone of the southern coastal economy, yet it faces the highest hurricane exposure in the nation ([FEMA, 2025a]; Fig 2). The region generates approximately $910 million annually in commercial dockside revenue (roughly 15% of the U.S. total) and lands the highest number of commercially harvested shrimp and oysters in the nation [NOAA Fisheries, 2024b]. The Gulf's recreational sector also contributes significantly to the U.S. recreational fishing economy, accounting for over one-third of the nation's fishing trip-related expenditures at $5.1 billion [NOAA Fisheries, 2024a]. Together, the Gulf's commercial and recreational fishing sectors support over 250,000 full- and part-time jobs [NOAA Fisheries, 2024a]. Beyond fisheries statistics, the Gulf also encompasses centuries-old cultural identities and critical food systems. The concentration of these fisheries in states such as Louisiana, Texas, and Florida creates a distinct vulnerability: the most productive fisheries in the Southeastern U.S. are located in the most storm-prone waters (Fig 2).

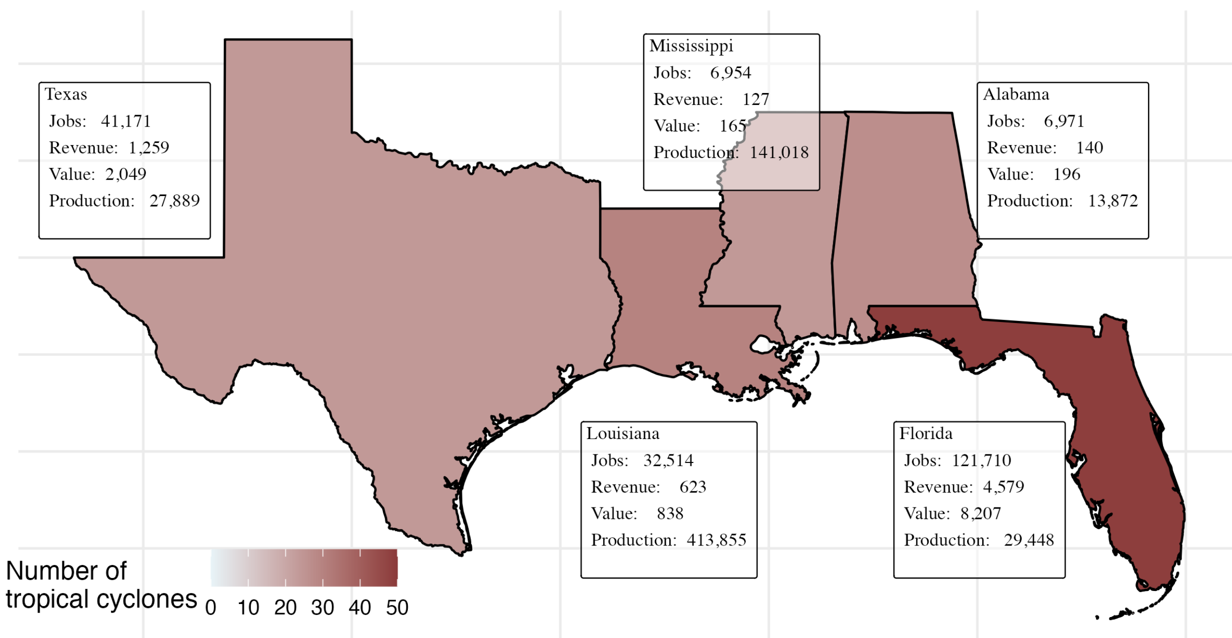

Figure 2: Commercial fisheries production and exposure to tropical cyclones in Gulf states. Colors in the map indicate the number of named storms that have made landfall in each state (2000 - 2025). The insets show fisheries socioeconomic statistics for jobs (number of jobs), revenue (Millions USD), value of the fishery sector as a whole (Millions USD), and seafood production figures (Tons) as reported by NOAA [NOAA Fisheries, 2024a].

Hurricanes and storms impose a double jeopardy on Gulf fisheries by simultaneously degrading the ecosystem and destroying physical capital. Ecologically, storms induce rapid salinity fluctuations, turbidity, and physical damage to habitats like seagrass and oyster beds, which in turn destabilize food webs and reduce biomass of heavily modified environments [Conner et al., 1989, Monnereau et al., 2015, Rothenberger et al., 2018]. Operationally, storm surge and wind may destroy critical coastal infrastructure such as marinas, icehouses, and processing facilities, as well as vessels, while effectively halting the labor force as crews focus on personal recovery [Ingles, 2007, Impact Assessment Inc, 2007, Caffey et al., 2019].

Lost catch from disruptions to environmental and operational conditions is costly, with existing estimates varying widely from \$100,000 to \$100M per storm [Caffey et al., 2007, Solís et al., 2013]. Damages to commercial seafood infrastructure can be as large as \$260,000 [Erlambang, 2008]. This combination of reduced yields and crippled capacity creates a feedback loop in which recovery can become slower and more expensive with each subsequent storm, thereby threatening the long-term viability and competitiveness of the sector.

In light of this risk, a complex web of institutions currently attempts to mitigate hurricane impacts. Federal support is bifurcated: NOAA focuses on forecasting and damage assessment, while FEMA and the Small Business Administration coordinate broader financial recovery through loans and disaster assistance [FEMA, 2025b]. State agencies, such as the Florida Fish and Wildlife Conservation Commission and the Louisiana Department of Wildlife and Fisheries, may bridge local gaps with emergency closures and state-level aid. However, even with this support, fishers indicate that governmental assistance can be administratively burdensome and slow to arrive [Ingles, 2007]. Consequently, recovery often depends on informal networks and NGOs rather than systemic resilience policies, which leaves the sector precariously exposed to future shocks.

Despite this evident vulnerability, the current federal trajectory presents mixed signals for resilience. The implementation of the Executive Order on Restoring American Seafood Competitiveness has primarily focused on deregulation, technology adoption, and streamlining NOAA's management to boost production and reduce the $20 billion seafood trade deficit. While this mandate calls for modernizing data collection and making management more responsive to real-time conditions, including environmental shocks, its core emphasis on reducing regulatory burdens and bypassing conservation efforts does not prioritize resilience. Concurrently, NOAA's ability to provide financial recovery has seen slight improvement following the Fisheries Disasters Improvement Act of 2022, which provided new criteria and timeliness for assistance. However, it is our assessment that these changes merely refine the reactive aid process, thereby falling short of implementing preventive strategies that would mitigate losses before disasters occur.

Taken together, these factors suggest that the current policy landscape is attempting to optimize the sector's output in pursuit of a competitive advantage, but without systematically addressing the increasing threat of climate-driven shocks, including hurricanes, that continue to undermine that output. This leaves Gulf fisheries in a delicate state where short-term efficiency gains can be erased by a single catastrophic event.

We note that this vulnerability extends to other shocks like marine heatwaves and harmful algal blooms. In the Pacific and Atlantic coasts, for instance, marine heatwaves have caused temporary redistribution of target species, leading to 40-60% reductions in biomass and catch [Chaikin et al., 2026, Free et al., 2023, Villaseñor Derbez et al., 2024]. In extreme cases, marine heatwaves can also induce total collapse of populations, as observed for snow crab in the Bering sea during 2021 [Szuwalski et al., 2023]. Harmful algal blooms also pose a serious threat to fisheries because high concentrations of toxic micro-algae can trigger fishery closures, leading to large economic losses [Jardine et al., 2020, Mao and Jardine, 2020]. Critically, fisheries that are more resistant to climate shocks such as marine heatwaves have also been shown to be more resistant to market shocks such as the COVID-19 pandemic [Villaseñor Derbez et al., 2025], suggesting that climate resilience confers broad cross-domain benefits.

Importantly, all of these shocks (i.e., hurricanes, marine heatwaves, and harmful algal blooms) are climate- and weather-related events with some degree of predictability [McGowan et al., 2017, Arcodia et al., 2025]. Accurate forecasts can help reduce damages by allowing for adaptation Molina and Rudik [forthcoming]. This feature highlights that reinforcing and expanding support for existing and new programs that monitor the evolution and prediction of these recurrent shocks could further contribute to the formation of robust and proactive fisheries management [Trainer et al., 2020].

From Relief to Industrial Strategy

To achieve the federal mandate of seafood competitiveness, strengthening FEMA and NOAA's capacity to predict and mitigate the impacts of climate shocks must be viewed as an act of industrial policy rather than simple disaster relief. Historically, the U.S. lags behind other developed nations in long-term investment in climate adaptation for coastal industries [Bakkensen and Mendelsohn, 2016]. We argue that closing this gap requires treating disaster management as strategic investment to stabilize and increase production when possible, thus fostering resilience. By viewing resilience as a prerequisite for market stability, the U.S. can ensure that the Gulf and other vulnerable regions remain as competitive pillars of the national seafood economy.

Achieving durable competitiveness also requires pairing physical infrastructure with robust information to guide decision-making under climate uncertainty. Currently, our understanding of the impacts of hurricanes and other climate shocks remains fragmented. Existing analyses are often focused on disparate parts of the supply chain and may fail to capture the integrated modern nature of the seafood trade, particularly at the national level. Advancing from reactive relief to proactive management requires a strategic commitment to decision-support science. This involves integrating high-resolution bio-economic models that treat fisheries not merely as biological stocks subject to management, but as sophisticated social-ecological systems that are part of a complex supply chain. Following this framework, scientific efforts should map the dependencies between stock status, fleet mobility, and fisheries infrastructure, as well as with processing and retail supply chains. With a stronger understanding of the entire value chain, from vessel to consumer, we can transform science into an actionable tool for decision-making.

Despite its strategic importance, there are reasons to believe that this scientific capacity is currently vulnerable. The recent suspension and restructuring of the Sixth National Climate Assessment (NCA) or the removal of federal climate data repositories represent a withdrawal of essential information that contributes to the sector's competitiveness. For example, the NCA was designed to provide the high-level insights that policymakers and stakeholders require to understand current environmental trends and volatility. Without the precise and timely assessments championed by the NCA and similar federal efforts, we conjecture, response to climate shocks is likely to remain reactive. Moving beyond this reactive framework and treating ecological monitoring and economic impact assessments as essential public capital presents a path toward a proactive management to secure seafood competitiveness.

Three Pillars to Support Competitive Seafood Markets

Here, we argue that securing the competitiveness of the Gulf seafood sector requires a multi-tiered policy approach. This approach is organized around three pillars: (1) robust and adaptive governance; (2) an independent safety net; and (3) enhanced clarity on systemic climate risk.

Pillar 1: Robust and adaptive governance. Clear and flexible rules are essential to supporting efficient and climate-resilient fisheries, which in turn provide the foundation for a competitive seafood market. State and federal rules govern who has access to natural resources, as well as how much and when they can extract. When designed properly, these rules can ensure well-functioning fisheries, prevent collapse, and support the rebuilding of fish stocks [Costello et al., 2008, Frank and Oremus, 2025]. Clarity reduces management complexity and facilitates timely guidance from managers and disaster relief agencies, while flexibility allows fishers and managers to adapt to dynamic climate conditions and better withstand environmental shocks. For example, fishers may adapt by modifying their gear or adjusting the timing and location of fishing [Ilosvay et al., 2022, Mason et al., 2022], while managers can adjust spatial or temporal closures, issue temporary permits to target alternative species, or modify—both downward and upward—total allowable catch limits [Mason et al., 2022]. More broadly, flexibility implies avoiding "one-size-fits-all" approaches that fail to account for the diversity of interactions in fisheries as complex social-ecological systems [Mahon et al., 2008].

Along this dimension, the U.S. already has a robust and adaptive approach to fisheries management [Frank and Oremus, 2025]. However, its stability may be threatened by ongoing executive actions aimed at reducing regulatory burdens under the premise that current levels of fisheries production are constrained by existing regulations. In contrast, ample evidence indicates that production is primarily constrained by economic incentives, or the lack thereof [Oremus et al., 2023]. Efforts to enhance American seafood competitiveness should focus on reinforcing clear and flexible rules, whether already in place or still to be developed, and on simplifying existing frameworks without compromising future harvests or catch stability in pursuit of short-term increases in yield.

Pillar 2: Independent safety net. While regulations can help mitigate the impacts of environmental shocks, they cannot prevent them altogether. As such, an independent safety net can enhance resilience when such shocks inevitably occur. To achieve this goal, the sector requires a politically insulated network of financial and logistical support mechanisms. This includes a) expanding access to affordable index-based risk insurance to reduce reliance on slow-moving federal aid; and b) streamlining Community Development Block Grant–Disaster Recovery funds. Index-based insurance schemes have long been used in agricultural production, where they have been shown to buffer the impacts of drought and, in some cases, reduce social conflict over shared resources [Tadesse et al., 2015, Gehring and Schaudt, 2026]. Other hurricane-prone regions, such as Saint Lucia and Grenada, have recently begun offering index-based insurance to their fishing sectors [Sainsbury et al., 2019]. Post-disaster recovery financial aid exists but is limited in scope. Future efforts should prioritize "Build Back Better" grants that induce adaptation and reduce the sector's exposure to hazards (e.g., development of storm-resistant cold storage and fuel systems). To be effective, we *emphasize* that these mechanisms must provide reliable protection against environmental risk, free from political/partisan disruption.

Pillar 3: Enhanced clarity on systemic climate risk. Finally, we argue that the sector requires a clear understanding of, and effective communication about, systemic climate risk. A central component of this pillar is systematically learning from past impacts. Therefore, policymakers should prioritize the development and dissemination of science-driven, quantitative assessments of hurricanes and other climate-related exposures [Link et al., 2015, Parmesan et al., 2022]. These assessments should comprehensively cover the industry, from stock health and fleet operations to infrastructure and distribution networks, and be paired with a transparent menu of mitigation and adaptation options tailored to fisheries and coastal communities.

Together, these pillars form an integrated framework for building a competitive and resilient American seafood industry. By reinforcing governance, stabilizing risk exposure, and improving information on climate threats, short-term efficiency gains can be sustained even as climate risks intensify.

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Data Availability Statement

All data and code to produce figures 1 and 2 is available at https://github.com/jcvdav/hurricanes_resilient_fisheries.

Conflict of Interest Statement

The authors declare no conflicts of interest.

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